Landslide Modelling Using Remote Sensing and GIS

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Abstract - **One of the forms of natural disasters given much attention by researchers all over the world during the International Decade for Natural Disaster Reduction (1990-2000) is the occurrence of landslides. Landslides take place in various regions of India, notably in the Himalayas and the Western Ghats, especially during monsoon. This paper describes some of the on-going studies related to modelling and risk evaluation of landslides at IIT Bombay, India. The studies described are: (i) rainfall induced landslide evaluation using DEM and GIS, (ii) fractal modelling of mass movement including critical failure surface determination using shortest path concept of GIS and (iii) evaluation of risk due to landslide. The unconventional methods used here demonstrate the usefulness of a GIS based technique in identifying potential slide regions in a long stretch of a hill slope and the corresponding risk, which could serve as supports for landslide hazard management.**

I. Introduction

Landslides are wide spread in several parts of India, notably in the Himalayas and the Western Ghats. This paper describes some of the on-going studies related to mathematical modelling of landslides at IIT Bombay. There are at least three stages in a landslide event, which deserve attention: determination of an admissible critical slip surface, prediction of the run-out and deposition profile of the debris and, finally, evaluation of the risk associated with the landslide. This paper addresses these three aspects. It attempts to combine the methodologies used for quantitative geotechnical and qualitative geomorphological slope stability evaluation using the concepts of Digital Elevation Modelling (DEM) and Geographic Information Systems (GIS) in three steps: (i) rainfall induced landslide evaluation using DEM and GIS (ii) fractal modelling of mass movement including critical failure surface determination using shortest path concept of GIS and (iii) evaluation of risk due to landslides.

II. Rain Induced Landslides

Occurrence of landslides along long stretches of hill roads is a common sight during heavy monsoon. This paper attempts to analyse the stability of a long stretch of a hill slope using DEM and GIS concepts. The case study of a landslide, which occurred on June 28, 1994 in the village of Parmachi (Latitude 18^0 8' N and Longitude 73^0 36' E) in the Western Ghats is considered. The landslide had occurred as a result of a heavy downpour of about 240 mm/day (0.1 m/hr) of rainfall. The methodology adopted here involves building a DEM first. For this, the study area of about 500 m x 400 m extent has been divided into grids of 16 m x 16 m size. The hill where the landslide occurred has elevations varying from about 240 m to 500 m. The slopes vary from 50^0 to 60^0. The soil is residual and the cover varies from 0.5 m to as much as 10 m.

A stability analysis has been carried out for different intensities and durations of rainfall [2] using the infinite slope method. This method has been used since it allows superposition of thematic information, especially for subsequent risk evaluation. A factor of safety value is obtained for each grid using (1).

$$F=\frac{c_d+\left[(\gamma_{sat}-\gamma_w)(h-z)+\gamma_1(H-h)\right]\cos^2\beta\tan\phi_d}{\left[\gamma_{sat}(h-z)+\gamma_1(H-h)\right]\cos\beta\sin\beta} \qquad (1)$$

where, c = effective soil cohesion (kN/m^2); γ_{sat}=saturated unit weight of the soil (kN/m^3); γ_1 = unit weight of the soil (kN/m^3); H = thickness of soil cover (m); h = groundwater table height (m); z = depth of the failure surface from the top (m); ϕ_d = angle of friction (degrees); β = inclination (degrees).

The fluctuation of the groundwater table, the effect of vegetation, in the form of root cohesion c_t and tree weight T are incorporated. The thematic information required for this is obtained from different sources. Remote sensing is useful in obtaining information about vegetation and landuse and the variable soil conditions. The result (Fig.1) clearly shows that a rainfall of 240 mm/day (0.01 m/hr) leads to a large patch of contiguous grids failing, indicating failure of the slope, i.e. landslide.

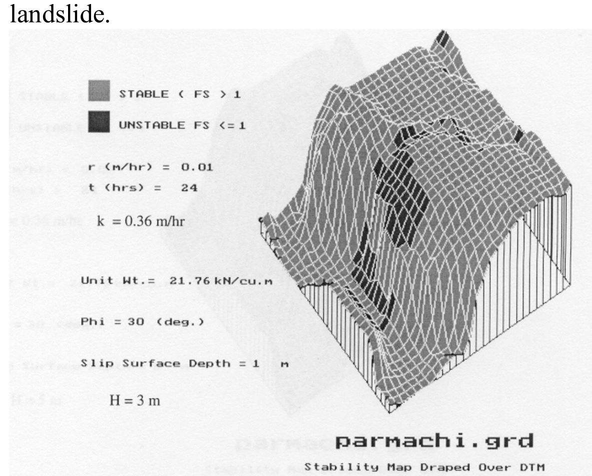

Fig. 1. Simulation of Parmachi landslide

III. Runout And Deposition

The run-out of a typical debris flow during a landslide depends on momentum transfer. In a landslide the sliding mass continuously decreases as deposition progresses. Here a fractal deposition model is evolved and used.

Deposition Pattern with Distance

The deposited mass M_d and distance travelled along the paths have a definite relationship. Two non-dimensional terms are calculated: (i) the term M_p/M_i, where M_i is the initial displaced mass and Mp is mass participating in movement (=1-M_d) at a location at distance **s** and (ii) the term $(s_f - s)/s_f$. A power law relationship for deposition pattern is derived from known deposition profiles of a number of landslides [2].

To compute M_i, the critical slip surface is located by the concept of shortest path, which is widely used in GIS network problems, for instance, in route location problems [3]. The critical slip surface will be analogous to the 'shortest path', i.e. the path with the least 'weighted' distance. The least distance, in this case, would mean the least factor of safety. Minimization of the well-known expression for overall factor of safety [4] gives the critical slip surface.

$$F = \frac{1}{L}\int F_L \, dL = \frac{1}{L}\Sigma\left\{\frac{c+(\sigma-u)\tan\phi}{\tau}\right\} b\sec\alpha \tag{2}$$

where F_L is the local factor of safety and is given by the term inside the brackets. σ and τ are the normal and shear stresses prevalent along a plane with α as the inclination with the horizontal; c, ϕ and u are the cohesion, angle of shearing resistance and pore pressure; b is the width of slices; and L is the total length of the slip surface. The term inside the summation sign in (2) is a weighted sum of local factors of safety, the weightage for each slice being the base length b sec α of the slice concerned.

The following relations are now derived in terms of the fractal dimension of the terrain from fundamental principles of mechanics [2]:

For run-out distance:

$$s_f = -\frac{v_0^2}{A}\{2\times(2.6479 - 1.15437\times D)+1\} + s_1 \tag{3}$$

where A = 2g (sinβ -μcosβ)

For deposition profile:

$$\frac{dM}{ds} = -\frac{M_i}{s_f}\times 1.4\times(2.6479 - 1.15437\times D)\times\left(\frac{s_f - s}{s_f}\right)^{1.6479-1.15437\times D} \tag{4}$$

A case study of the Madison Canyon Landslide, which occurred on August 17, 1959 in the mountain terrain of south western Montana on Madison river in North America is used for validation. The displaced mass moved down about 2200 m and travelled about 600 m, crossed the river and rode up the opposite bank. The computed deposition profile is compared with the observed in Table 1. After having validated the fractal model, the deposition profile of the Parmachi landslide is computed and presented in Table 2. This agrees well with the observed values. The volume of displaced soil and rock was approximately 30,000 m^3 and the run-out was about 300 m.

TABLE I
Deposition Profile

Location	Dist. (m)	Thickness of deposit (m)	
		Reported	Computed
3	240	0	0
6	440	36	77
7	520	68	71
8	600	65	66
9	680	69	61
10	760	68	55
11	840	53	46
s_f	912	--	0

TABLE II
Deposition Profile for Parmachi

Distance (m)	Moving Mass (cu. m)	Thickness of Deposition (m)
100	294.19	0
150	206.38	1.75
200	127.96	1.57
250	61.25	1.33
275	33.62	1.11
300	11.37	0.89
321	0	0

IV. Risk Evaluation

For slope stability problems Risk may be defined as:

Risk = Probability of occurrence of critical rainfall x
Probability of occurrence of slope failure x
Consequences of failure

The probability of failure is calculated based on the consideration that the critical surface corresponds to the minimum value of the conventional factor of safety. If the latter is denoted by S_c (i.e. S_c is the critical surface for which FS is minimum) and the probability of its occurrence by $P[S_c]$, then the probability of failure p_f is written as

$$p_f = P[F \mid S_c]\, P[S_c] + \sum_{i=1}^{N-1} P[F \mid S_i] P[S_i] \quad (5)$$

Here, the expected value and standard deviation of FS, which are required for the computation of p_f are obtained using Rosenblueth's Point Estimate Method [5]. The computations can be further improved if fuzzy uncertainties are also considered in the analysis [6].

TABLE III
DECISION TABLE

Chance Node	Event	Event Probability
1	Critical rainfall of 240 mm/day	0.70
	No rainfall	0.30
2	Landslide	0.467
	No landslide	0.533
3	a) Damage to structures	0.70
	b) Loss of life	0.20
	c) Other losses	0.10

Table 3 shows the probabilities of all likely events for a critical rainfall of 240 mm/day (0.01 m/hr).The alternative with the least expected cost is chosen if the expected value is the criterion for decision. Thus the evaluation of risk could help in deciding and planning suitable remedial or preventive measures.

V. CONCLUSIONS

The proposed methods are useful in evaluating occurrence of landslides along long stretches of hill slopes. The concept of GIS can be used in three different contexts in a landslide problem. Firstly, the shortest path method is useful in determining the critical slip surface. The DEM and raster GIS based approach is useful in determining the regions of the slope, which are likely to slide. The information about landuse is useful in risk evaluation. The computation of risk can be improved further by considering fuzzy uncertainties as well in the formulation. The interpretation of the risk probabilities, if done with care and insight, could serve as instrument for anticipating and planning mitigation measures.

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